PHASE 4 — THE ENTERPRISE CRYPTOGRAPHIC ECOSYSTEM

Project: Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

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---

1. Purpose of This Phase

The first three phases established:

```
Phase 1
Cryptography
        ↓
Phase 2
Quantum Threat
        ↓
```

```
    Phase 3
    Post-Quantum Cryptography
```

Now we face the real engineering problem:

> **Where is all this cryptography actually being used?**

An enterprise does not have one cryptographic system.

It may have thousands.

Cryptography exists inside:

- Source code
- Libraries
- Containers
- Operating systems
- Network protocols
- Certificates
- Cloud services
- Hardware
- Databases
- Mobile applications
- Firmware
- Third-party software
- CI/CD infrastructure

Therefore, ECDAT cannot be a simple source-code scanner.

It must become an **enterprise-wide cryptographic discovery system**.

---

2. The Central Problem: Cryptography Is Everywhere

Consider a hypothetical organization.

```
Enterprise
│
├── 3,000 Repositories
├── 15,000 Applications
├── 40,000 Containers
├── 200,000 Certificates
├── 100,000 Keys
├── 15 Cloud Accounts
├── 5,000 Servers
├── 10,000 Network Devices
├── 2,000 APIs
└── 50,000 Employees
```

Cryptographic operations may exist in every layer.

A developer might write:

```
AES.new(key, AES.MODE_GCM)
```

A Java service might use:

```
Cipher.getInstance("AES/GCM/NoPadding");
```

A server might automatically negotiate:

```
TLS 1.3
ECDHE
AES-256-GCM
```

A cloud service might use:

```
AWS KMS
```

A Kubernetes cluster might use:

```
TLS certificates
```

A firmware image might contain:

```
RSA public key
```

The cryptography may not be visible from a single place.

That is the core discovery challenge.

---

3. What Is a Cryptographic Asset?

ECDAT needs a precise definition.

A **cryptographic asset** is any software, hardware, configuration, key, certificate, protocol, dependency, service, or other resource that performs, stores, references, manages, or depends upon cryptographic operations.

Examples:

```
Algorithm
Key
Certificate
Library
API
Protocol
HSM
KMS
Crypto Provider
Configuration
Firmware
Container
Application
```

---

4. Cryptographic Asset Taxonomy

ECDAT can organize assets into major categories.

```
Cryptographic Assets
│
├── Algorithms
│
├── Keys
│
├── Certificates
│
├── Protocols
```

```
      │
      ├── Libraries
      │
      ├── APIs
      │
      ├── Applications
      │
      ├── Hardware
```

Each asset can then have relationships.

For example:

```
Application
   │
   ├── uses → OpenSSL
   │
   ├── uses → RSA
   │
   ├── presents → Certificate
   │
   └── protects → Customer Data
```

---

5. Source Code

Source code is one of the most important discovery targets.

Developers may directly invoke cryptographic APIs.

Example:

```
from cryptography.hazmat.primitives.ciphers import Cipher
```

Or:

```
crypto.createCipheriv(...)
```

Or:

```
Cipher.getInstance(...)
```

Or:

```
aes.NewCipher(...)
```

ECDAT should detect these patterns.

But this is only the beginning.

---

6. Programming Languages

Enterprise environments use many languages.

Potentially relevant languages include:

```
Python
Java
JavaScript
TypeScript
```

```
C
C++
C#
Go
Rust
Swift
Kotlin
PHP
Ruby
Dart
Scala
Objective-C
```

Each language has different cryptographic libraries and APIs.

Therefore, ECDAT needs language-specific detection logic.

---

7. Cryptographic APIs

Applications rarely implement cryptography mathematically from scratch.

Instead they use APIs.

Example:

```
Application
    ↓
Crypto API
    ↓
```

```
Crypto Library
    ↓
Algorithm
```

For example:

```
Java Application
       ↓
JCA/JCE
       ↓
Provider
       ↓
OpenSSL / Native Crypto
```

This creates a dependency chain.

ECDAT needs to understand the entire chain.

---

8. Cryptographic Libraries

Common cryptographic libraries include:

- OpenSSL
- BoringSSL
- LibreSSL
- libsodium
- Bouncy Castle
- Botan

- wolfSSL
- mbed TLS
- Java security providers
- Windows CNG

Libraries are especially important because one vulnerable library can affect hundreds of applications.

---

9. OpenSSL

OpenSSL is one of the most widely deployed cryptographic libraries.

It supports:

- TLS
- X.509
- RSA
- ECC
- AES
- SHA
- HMAC
- Certificates
- Key management

ECDAT should identify:

```
OpenSSL
Version
Build
```

```
Configuration
Applications
Algorithms
```

For example:

```
OpenSSL 3.x
        ↓
TLS
        ↓
ECDHE
        ↓
AES-GCM
```

A library-level dependency may be more important than a single source-code occurrence.

---

10. BoringSSL

BoringSSL is Google's fork of OpenSSL used in major systems.

It is used in areas including:

- Chrome
- Android
- Google infrastructure

ECDAT should not assume that:

```
OpenSSL detected
```

means every OpenSSL-like implementation is actually OpenSSL.

Library identity matters.

---

11. LibreSSL

LibreSSL is another OpenSSL-derived project.

It is used in various operating systems and applications.

ECDAT should maintain separate library identities.

---

12. Java Cryptography Architecture

Java applications often use the:

Java Cryptography Architecture (JCA)

and:

Java Cryptography Extension (JCE)

Applications may request:

```
Cipher.getInstance(...)
```

or:

```
Signature.getInstance(...)
```

The actual implementation can depend on the configured provider.

This creates an important discovery problem:

```
Java API
   ↓
Provider
   ↓
Implementation
   ↓
Algorithm
```

A regex scanner may identify `Cipher`, but determining the actual cryptographic primitive may require deeper analysis.

---

13. .NET Cryptography

.NET provides APIs such as:

```
System.Security.Cryptography
```

Examples include:

- RSA
- ECDsa
- ECDiffieHellman
- AES
- SHA
- HMAC

ECDAT should understand both the API and the implementation context.

---

14. Python Cryptography

Python applications may use:

```
cryptography
PyCryptodome
hashlib
ssl
OpenSSL bindings
```

Example:

```
AES.new(...)
```

But a sophisticated scanner must distinguish:

```
Hashing for integrity
```

from:

```
Hashing used as a cryptographic security mechanism
```

Context matters.

---

15. Go Cryptography

Go has a strong standard cryptographic ecosystem.

Examples:

```
crypto/aes
crypto/rsa
crypto/ecdsa
crypto/ed25519
crypto/tls
crypto/sha256
```

Go's standard library makes static discovery relatively structured.

ECDAT should still identify:

- Algorithm
- Key size
- Usage
- TLS context
- Certificate usage

---

16. Rust Cryptography

Rust has multiple cryptographic ecosystems.

Examples include:

```
RustCrypto
ring
rustls
OpenSSL bindings
```

Dependency analysis becomes important because Rust projects may use different cryptographic implementations.

---

17. Node.js Cryptography

Node.js provides:

```
node:crypto
```

Applications may use:

- AES
- RSA
- ECDSA
- Diffie-Hellman
- HMAC
- Hashes
- TLS

ECDAT should detect both direct API calls and dependencies.

---

18. Native and System Libraries

Applications may dynamically link to:

```
libcrypto
libssl
```

or other system cryptographic libraries.

The source code may contain no explicit implementation.

Example:

```
Application
      ↓
Dynamic Linker
      ↓
libcrypto.so
      ↓
OpenSSL
```

Therefore, binary analysis becomes essential.

---

19. Static and Dynamic Linking

There are two major linking models.

Dynamic Linking

```
Application
      ↓
Shared Library
```

The cryptographic implementation exists externally.

Static Linking

```
Application
      ↓
Embedded Library Code
```

The cryptographic implementation may be inside the binary.

This makes discovery harder.

ECDAT therefore needs binary inspection capabilities.

---

20. Package Managers and Dependencies

Modern applications depend heavily on packages.

Examples:

```
npm
pip
Maven
Gradle
NuGet
Cargo
Go Modules
```

A developer may not directly call OpenSSL.

Instead:

```
Application
 ↓
```

```
Framework
 ↓
Dependency
 ↓
Crypto Library
 ↓
Algorithm
```

Therefore, dependency graphs are essential.

---

21. Source Repositories

The NTRO problem specifically mentions scanning source-code repositories.

Possible sources include:

- GitHub
- GitLab
- Bitbucket
- Internal Git servers
- Monorepos
- Archive repositories

ECDAT should support:

```
Repository
↓
Branch
```

```
↓
Commit
↓
Files
↓
Code
↓
Dependencies
↓
Crypto Assets
```

---

22. Configuration Files

Cryptography is often configured rather than coded.

Examples:

```
nginx.conf
application.yml
server.xml
openssl.cnf
ssh_config
```

Configuration may specify:

```
TLS versions
Cipher suites
Certificate paths
```

```
Key paths
Signature algorithms
```

Therefore, configuration scanning is essential.

---

23. Environment Variables

Applications may load cryptographic configuration through environment variables.

Example:

```
TLS_CERT=/certs/server.pem
TLS_KEY=/secrets/server.key
```

or:

```
KMS_KEY_ID=...
```

ECDAT should detect references without exposing sensitive secret values.

---

24. Secrets and Key Material

This is a critical distinction.

ECDAT should discover **cryptographic keys and key references**, but it should not become a secret-exfiltration tool.

For example:

```
PRIVATE_KEY=/vault/prod/key
```

should be represented as:

```
Key Reference:
Vault path

Algorithm:
Unknown

Location:
Production API

Secret Value:
NOT COLLECTED
```

The platform should follow least-privilege principles.

---

25. Digital Certificates

Certificates are among the easiest cryptographic assets to inventory.

ECDAT can extract:

```
Subject
Issuer
Serial Number
Validity
Public Key Algorithm
Key Size
Signature Algorithm
```

```
SAN
Certificate Chain
```

Example:

```
Certificate
│
├── Algorithm: RSA
├── Key Size: 2048
├── Issuer: Example CA
├── Valid Until: ...
└── Application: Payments API
```

---

26. TLS Infrastructure

TLS is one of the largest cryptographic attack surfaces.

ECDAT should identify:

```
TLS Version
Cipher Suite
Key Exchange
Certificate Algorithm
Signature Algorithm
Crypto Library
Server
Endpoint
```

For example:

```
TLS 1.3
+
ECDHE
+
AES-256-GCM
+
ECDSA P-256 Certificate
```

This is a richer asset than simply saying:

```
ECC found
```

---

27. SSH Infrastructure

SSH may use:

- RSA
- ECDSA
- Ed25519
- Diffie-Hellman
- MAC algorithms
- Symmetric encryption

ECDAT should inventory:

```
SSH Server
Host Key
Host Key Algorithm
KEX Algorithm
Cipher
MAC
Configuration
```

---

28. VPN Infrastructure

VPN systems can use cryptography for:

- Authentication
- Key exchange
- Encryption
- Integrity

Examples:

- IPsec
- WireGuard
- OpenVPN
- SSL VPN

ECDAT should identify cryptographic mechanisms at each layer.

---

29. PKI Infrastructure

Enterprise PKI may contain:

```
Root CA
│
├── Intermediate CA
│      │
│      ├── Service Certificates
│      ├── User Certificates
│      └── Device Certificates
│
└── Code Signing CA
```

A change to the cryptographic algorithm can affect the entire hierarchy.

This is why PKI should be represented as a graph.

---

30. Hardware Security Modules

HSMs provide hardware-backed cryptographic protection.

They may store:

- Private keys
- Signing keys
- Encryption keys
- Certificate authority keys

ECDAT should identify:

```
HSM
 |
 ├── Vendor
 ├── Model
 ├── Firmware
 ├── Algorithms
 ├── Key Types
 ├── Applications
 └── PQC Support
```

---

31. Cloud Key Management

Cloud KMS platforms are critical.

Examples include:

- AWS KMS
- Azure Key Vault
- Google Cloud KMS

Applications may never see the underlying cryptographic keys.

Instead:

```
Application
     ↓
Cloud KMS API
     ↓
Managed Key
```

```
        ↓
Cryptographic Operation
```

Therefore, source-code scanning alone cannot discover the complete crypto dependency.

---

32. Cloud Certificate Services

Cloud environments may automatically issue and rotate certificates.

ECDAT needs to understand:

```
Certificate Service
      ↓
Certificate
      ↓
Application
      ↓
Load Balancer
      ↓
TLS
```

This is particularly important because certificate lifecycle can be completely automated.

---

33. Databases

Cryptography may exist at multiple levels.

Application-level encryption

```
Application
↓
Encrypt
↓
Database
```

Database-level encryption

```
Database
↓
Transparent Encryption
```

Storage-level encryption

```
Disk
↓
Encryption
```

ECDAT should distinguish these.

---

34. Storage Systems

Examples:

- Object storage
- File systems
- Block storage

- Backup storage
- Archives

Cryptography may be used for:

- Encryption at rest
- Integrity
- Access control

The important question becomes:

> What algorithm protects the data, and who controls the key?

---

35. Backups and Archives

Long-term archives are especially important for post-quantum risk.

Example:

```
Backup
Retention:
25 years

Encryption:
RSA-based key wrapping

Risk:
Potential HNDL exposure
```

Even if the application itself is modern, old backups may retain vulnerable cryptographic protection.

ECDAT should therefore include historical data stores.

---

36. API Gateways

API gateways frequently terminate TLS.

Architecture:

```
Internet
   ↓
API Gateway
   ↓
Microservice
```

The cryptographic dependency may therefore exist entirely at the gateway.

ECDAT must map:

```
Gateway
↓
Certificate
↓
TLS
↓
Crypto Algorithm
↓
Backend Service
```

---

37. Service Meshes

Modern microservice architectures often use service meshes.

Examples include:

- Istio
- Linkerd

They may automatically provide:

```
mTLS
```

between services.

Therefore, developers may write:

```
No cryptographic code
```

while the system still uses extensive cryptography.

This is a perfect example of why source scanning alone is insufficient.

---

38. Containers

Containers introduce another discovery surface.

An application container may contain:

```
Application
+
Libraries
```

```
+
OS Packages
+
Certificates
+
Crypto Libraries
```

ECDAT should scan:

```
Dockerfile
Image Layers
Packages
Libraries
Binaries
Certificates
Configurations
```

---

39. Docker Images

A Docker image is composed of layers.

Example:

```
Image
│
├── Base OS
├── System Libraries
├── Application Dependencies
```

```
    ├── Application
    └── Configuration
```

A cryptographic library may exist in the base image even if the application doesn't explicitly install it.

ECDAT should therefore track dependency provenance.

---

40. Kubernetes

Kubernetes introduces:

- Secrets
- ConfigMaps
- Ingress
- Service Mesh
- TLS
- Certificates
- Controllers
- Admission systems

A Kubernetes cluster can contain enormous numbers of cryptographic dependencies.

ECDAT should understand:

```
Cluster
↓
Namespace
↓
Workload
```

```
↓
Service
↓
Ingress
↓
Certificate
↓
Crypto Algorithm
```

---

41. CI/CD Pipelines

Cryptography also exists in build systems.

Examples:

```
Git signing
SSH keys
TLS
Package signing
Artifact signing
Secrets
Cloud credentials
```

A CI/CD pipeline may therefore have cryptographic dependencies that are invisible from application repositories.

---

42. Software Supply Chain

The software supply chain includes:

```
Developer
↓
Repository
↓
Build
↓
Dependency Resolution
↓
Artifact
↓
Signing
↓
Registry
↓
Deployment
```

Each stage can involve cryptography.

ECDAT should eventually connect these stages.

---

43. Code Signing

Code signing is particularly sensitive.

A private signing key allows an organization to establish trust in software artifacts.

If a signing system uses quantum-vulnerable public-key cryptography, migration becomes strategically important.

Potential assets:

```
Signing Key
Certificate
Signing Algorithm
Build Server
Signing Service
HSM
Artifact Registry
```

---

44. Firmware

Firmware can contain:

- Public keys
- Signature verification code
- Encryption
- Secure boot logic

For example:

```
Boot ROM
   ↓
Bootloader
   ↓
Firmware Signature
   ↓
Verify Public Key
```

```
        ↓
    Run Firmware
```

If the trust anchor is quantum-vulnerable, future migration can be difficult.

---

45. IoT and Embedded Systems

IoT systems may use:

- Device certificates
- Secure boot
- TLS
- Firmware signatures
- Device keys

Challenges include:

- Limited CPU
- Limited memory
- Long deployment lifetime
- Infrequent updates
- Physical deployment

These factors increase migration complexity.

---

46. Mobile Applications

Mobile applications may include:

- Certificate pinning
- TLS
- Local encryption
- Secure storage
- Digital signatures
- Cryptographic libraries

ECDAT could scan:

```
APK
IPA
Native Libraries
Embedded Certificates
Embedded Keys
Crypto APIs
```

---

47. Blockchain Systems

Blockchain systems may use:

- Digital signatures
- Hash functions
- Merkle trees
- Consensus cryptography
- Wallet keys

ECDAT could classify:

```
Blockchain
│
├── Signature Scheme
├── Hash Function
├── Wallet Keys
├── Smart Contracts
└── Cryptographic Dependencies
```

This is particularly relevant to the SIH theme.

---

48. Third-Party and SaaS Dependencies

An organization may rely on:

```
Payment Gateway
Identity Provider
Cloud Provider
SaaS Platform
External API
```

The organization may not control the underlying cryptography.

ECDAT should therefore distinguish:

```
Owned Crypto
```

from:

```
Third-Party Crypto
```

and:

```
Unknown Crypto
```

This becomes important for migration planning.

---

49. Legacy Systems

Legacy systems represent some of the hardest migration problems.

Example:

```
Mainframe
   ↓
Legacy Application
   ↓
RSA
   ↓
Custom Hardware
   ↓
No Modern API
```

Such systems may have:

- undocumented cryptography
- custom implementations
- unsupported libraries

- hardcoded keys
- no source code
- long replacement cycles

ECDAT should assign higher uncertainty and migration complexity to these systems.

---

50. Network Infrastructure

Network devices can contain:

- TLS
- SSH
- IPsec
- Certificates
- Device keys
- Firmware signatures

Examples:

- Routers
- Firewalls
- Load balancers
- Proxies
- IDS/IPS
- VPN gateways

This expands ECDAT beyond application security into infrastructure security.

---

51. Cryptographic Discovery Challenges

Now we reach one of the most important engineering sections.

Finding cryptography is surprisingly difficult.

Why?

Because cryptographic usage exists in several forms.

---

52. Direct vs Indirect Cryptographic Usage

Direct

```
AES.new(...)
```

Easy to detect.

Indirect

```
secure_store.encrypt(...)
```

The actual algorithm is hidden behind another abstraction.

Deep Indirect

```
Application
↓
Framework
```

```
↓
Dependency
↓
Provider
↓
Library
↓
Algorithm
```

This requires dependency analysis and semantic reasoning.

---

53. Explicit vs Implicit Cryptography

Explicit:

```
RSA
ECDSA
AES
```

Implicit:

```
HTTPS
mTLS
SSH
KMS
Secure Boot
Signed Package
```

The system may never mention "RSA" directly.

Yet cryptography is still present.

Therefore, ECDAT should discover **cryptographic behavior**, not merely cryptographic keywords.

---

54. Static vs Runtime Discovery

Static analysis examines:

```
Source Code
Binary
Container
Configuration
Dependencies
```

Runtime analysis examines:

```
Live Traffic
Processes
System Calls
TLS Handshakes
Loaded Libraries
```

Both provide different information.

Static Analysis

Advantages:

- Safe
- Repeatable
- Broad
- Works before deployment

Limitations:

- May miss dynamic behavior
- May overestimate unused dependencies

Runtime Analysis

Advantages:

- Shows actual behavior
- Detects dynamically loaded libraries
- Captures real protocol usage

Limitations:

- Requires deployment access
- Can miss rarely executed code
- More operationally complex

---

55. Building the Enterprise Crypto Map

ECDAT should construct a global model.

```
Enterprise
│
├── Business Unit
│      │
│      └── Application
│             │
│             ├── Repository
│             ├── Container
│             ├── Library
│             ├── Certificate
│             └── Crypto Algorithm
│
├── Infrastructure
│      │
│      ├── Server
│      ├── Firewall
│      ├── VPN
│      └── HSM
```

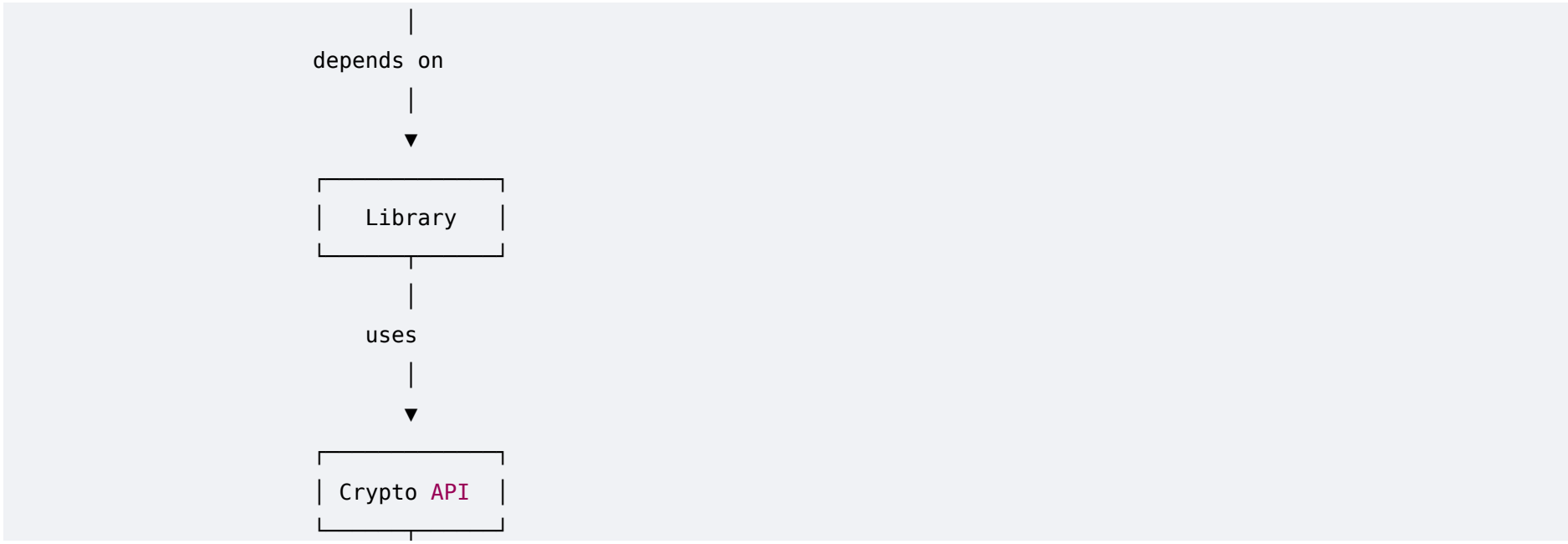
This becomes the foundation for the enterprise cryptographic inventory.

---

56. Cryptographic Dependency Graph

A graph representation could look like:

```
                    ┌─────────────┐
                    │ Application │
                    └──────┬──────┘
```



This graph becomes incredibly powerful when combined with quantum-risk information.

---

57. ECDAT Asset Model

A cryptographic asset could be represented as:

```
{
  "asset_id": "CRYPTO-001928",
  "asset_type": "algorithm",
  "algorithm": "ECDSA",
  "parameter_set": "P-256",
  "function": "digital_signature",
  "location": "payments-service",
```

```
        "repository": "payments-api",
        "library": "OpenSSL",
        "library_version": "3.x",
        "protocol": "TLS",
        "business_criticality": "critical",
        "data_classification": "restricted",
        "data_lifetime_years": 15,
        "quantum_status": "vulnerable",
        "migration_complexity": "high",
        "confidence": 0.96
```

This is the type of normalized record that could eventually feed the CBOM.

---

58. Discovery Confidence

Every discovery should have a confidence value.

Example:

```
Direct API Match
Confidence = 0.99
```

```
Binary Signature Match
Confidence = 0.93
```

```
Dependency Inference
Confidence = 0.82
```

```
LLM Semantic Inference
Confidence = 0.71
```

This prevents ECDAT from presenting uncertain findings as absolute facts.

---

59. Asset Ownership

Finding an asset is not enough.

The organization needs to know:

> Who is responsible for it?

Therefore:

```
Crypto Asset
      ↓
Application
      ↓
Team
      ↓
Business Owner
      ↓
Security Owner
```

This enables actionable migration planning.

Instead of:

> "4,000 RSA assets detected."

ECDAT should say:

> "Team Payments owns 381 high-priority RSA dependencies."

That is significantly more useful.

---

60. Data Classification

ECDAT should ideally associate cryptographic assets with data classifications.

For example:

```
Public
Internal
Confidential
Restricted
Highly Restricted
```

The exact taxonomy should be configurable.

This information directly affects quantum risk.

---

61. Business Context

Technical cryptography cannot be evaluated in isolation.

Consider:

```
Algorithm:
RSA-2048
```

This tells us very little.

Now add:

```
Application:
National Identity Service

Data:
Identity Records

Retention:
30 years

Business Criticality:
Critical

Exposure:
Internet

Migration:
5 years
```

Now ECDAT can produce a meaningful risk assessment.

---

62. Enterprise Crypto Inventory

The final inventory should look conceptually like:

Asset	Type	Algorithm	Location	Owner	Criticality	Quantum Risk
CRYPTO-001	Certificate	RSA-2048	API Gateway	Payments	Critical	High
CRYPTO-002	Key Exchange	ECDH P-256	VPN	Network	High	Critical
CRYPTO-003	Encryption	AES-256	Database	Data	Critical	Lower
CRYPTO-004	Signature	ECDSA P-256	CI/CD	DevOps	High	Critical
CRYPTO-005	Hash	SHA-256	Repository	Platform	Medium	Context-dependent

This is the beginning of the **Cryptographic Bill of Materials**.

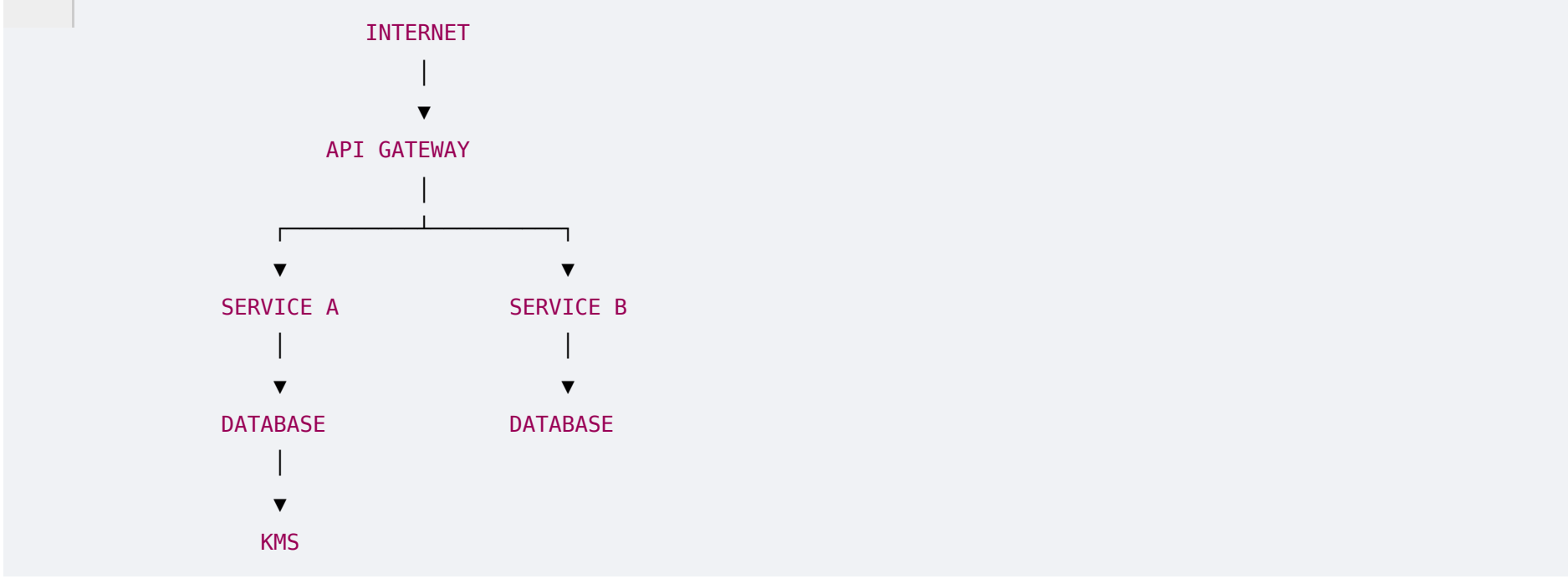
---

63. Example Enterprise

Let's imagine:

```
NATIONAL DIGITAL SERVICES
```

Infrastructure:



Additional infrastructure:

```
Cloud
|
├── Kubernetes
├── Load Balancers
├── KMS
```

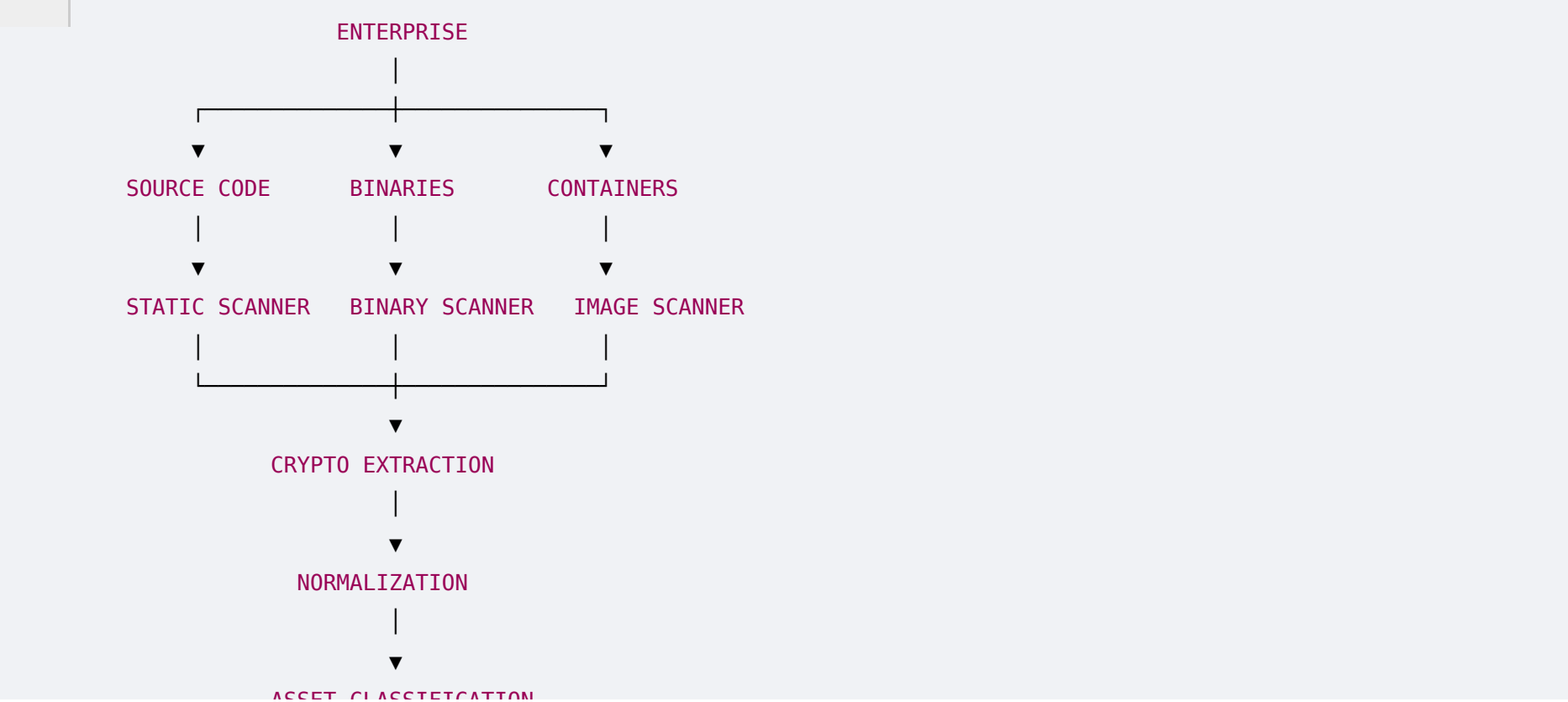
```
      ├── Certificates
      └── Secrets Manager
```

ECDAT should discover cryptography across the entire architecture.

---

64. ECDAT Discovery Pipeline

The enterprise discovery architecture can now be defined:



This architecture will become increasingly detailed in later phases.

65. Requirements Derived From This Phase

Phase 4 produces a major set of requirements.

R1 — Multi-source discovery

ECDAT must support multiple asset sources.

R2 — Multi-language source analysis

At minimum, the architecture should be extensible for major enterprise languages.

R3 — Library identification

The system must identify cryptographic libraries and versions.

R4 — Binary analysis

ECDAT must inspect binaries and linked libraries.

R5 — Container analysis

ECDAT must scan container images and layers.

R6 — Configuration analysis

Cryptographic settings must be discovered.

R7 — Certificate analysis

Certificates must be parsed and classified.

R8 — Cloud integration

Cloud KMS and certificate systems should be discoverable.

R9 — Runtime visibility

The architecture should eventually support runtime telemetry.

R10 — Dependency mapping

Cryptographic assets must be linked to applications and business owners.

R11 — Confidence scoring

Every finding should carry evidence and confidence.

R12 — No secret exfiltration

ECDAT must distinguish between:

```
Discovering a key exists
```

and:

```
Collecting the key itself
```

The latter should generally be avoided unless explicitly required and securely authorized.

---

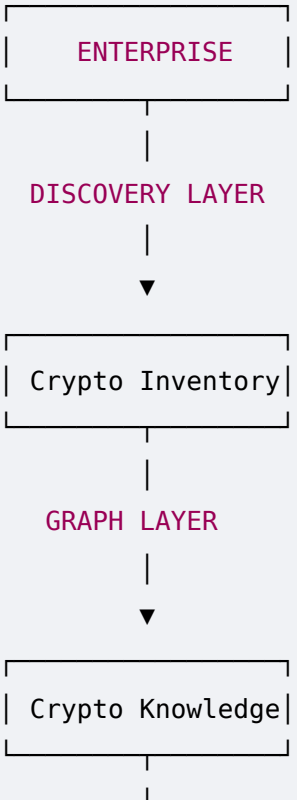
66. The Most Important Architectural Insight

After Phase 4, the ECDAT problem becomes much clearer.

The system isn't simply:

```
Scanner
```

It is:



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67. Phase 4 Summary

The enterprise cryptographic ecosystem is enormous.

Cryptography can exist in:

- Applications
- Source code
- Libraries
- Binaries
- Containers
- Kubernetes
- Certificates
- TLS
- SSH
- VPN
- PKI
- HSMs
- Cloud KMS
- Databases
- Backups
- APIs
- Service meshes
- CI/CD

- Code signing
- Firmware
- IoT
- Mobile applications
- Blockchain
- Third-party systems
- Legacy infrastructure

Therefore, ECDAT must discover **cryptographic behavior across the entire enterprise**, not merely search source code for algorithm names.

The key transition is:

```
Find Cryptography
       ↓
Understand Cryptography
       ↓
Map Dependencies
       ↓
Understand Business Context
       ↓
Assess Quantum Risk
       ↓
Recommend Migration
```

---

68. The Core Insight

The most important conclusion from this phase is:

> **Cryptographic discovery is an asset-management problem, a dependency-analysis problem, a cybersecurity problem, and ultimately a business-context problem.**

A simple scanner can tell an organization:

> "RSA exists."

A true ECDAT platform should be able to tell it:

> "RSA-2048 is used by the production payments gateway through OpenSSL, protects an internet-facing critical service, is associated with data retained for 15 years, depends on three infrastructure components, has a high migration complexity, and should be prioritized for PQC migration."

That is the difference between a **scanner** and a **cryptographic intelligence platform**.

---

69. Phase 5 Preview — Cryptographic Discovery & CBOM

Now we have reached one of the most important phases of the entire project.

Phase 5 will turn the enterprise ecosystem into an actual **discovery and inventory system**.

We will go deep into:

- What CBOM actually is
- CBOM vs SBOM
- Cryptographic asset schemas
- CycloneDX
- SPDX

- CryptoBOM concepts
- Algorithm detection
- API detection
- AST analysis
- Regex detection
- Dependency analysis
- Binary analysis
- Container analysis
- Certificate parsing
- Protocol discovery
- Library fingerprinting
- YARA-style detection
- Semgrep
- Tree-sitter
- CodeQL
- Trivy
- Syft
- Grype
- OpenSSL inspection
- Binary reverse engineering
- Evidence collection
- False positives

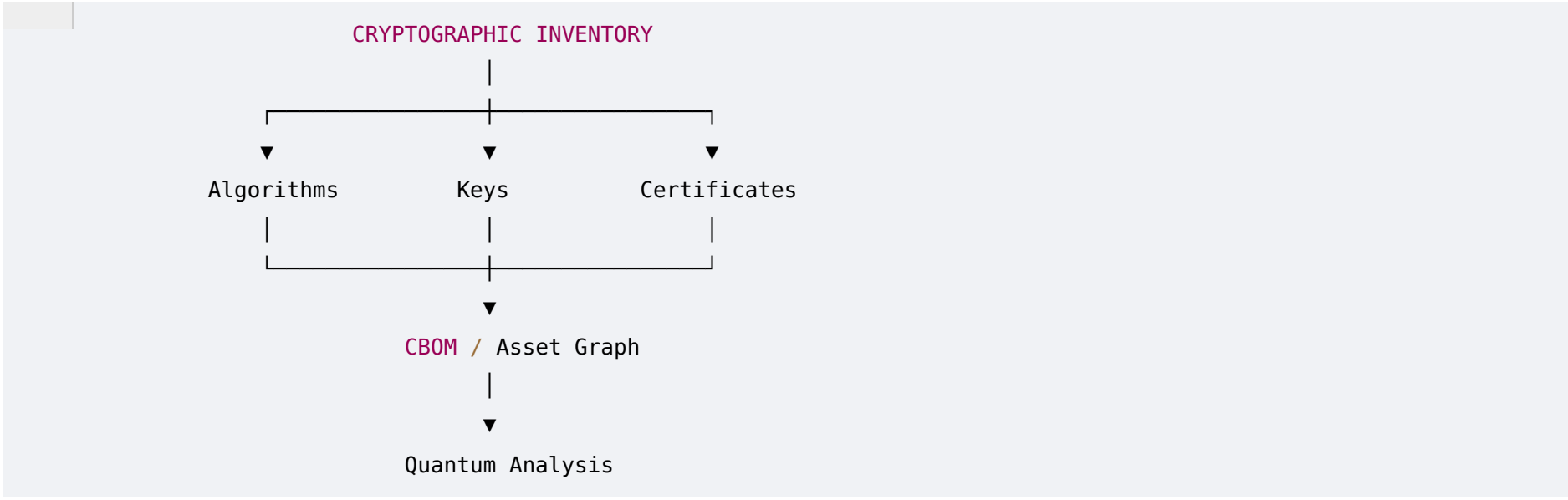
- False negatives
- Confidence scoring
- Asset deduplication
- Version tracking
- Crypto lineage
- CBOM generation
- CBOM storage
- CBOM visualization
- Incremental scanning
- Continuous discovery
- Git integration
- CI/CD integration

Most importantly, we'll design the **actual ECDAT discovery engine** that can take:

```
Repository
     +
Binary
     +
Library
     +
Container
     +
Certificate
```

```
        +
    Configuration
```

and transform it into:



Phase 4 complete.